

Домашнее задание №3

Анализ трафика в Wireshark

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1. Цели и задачи работы

2. Процесс выполнения лабораторной работы

1. 1. Цели и задачи работы

1.1 Цель лабораторной работы

Изучить MAC-адресацию и выполнить анализ сетевого трафика в Wireshark на канальном и транспортном уровнях.

1.2 Задачи лабораторной работы

- определить IP- и MAC-адреса сетевых интерфейсов;

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- изучить HTTP, DNS и QUIC-трафик;

1.2 Задачи лабораторной работы

- определить IP- и MAC-адреса сетевых интерфейсов;
- проанализировать кадры ARP и ICMP;
- изучить HTTP, DNS и QUIC-трафик;
- проанализировать TCP handshake.

2. 2. Процесс выполнения лабораторной работы

2.1 Определяю параметры сетевого подключения

```
Адаптер беспроводной локальной сети Беспроводная сеть:

DNS-суффикс подключения . . . . . :
Локальный IPv6-адрес канала . . . : fe80::da3b:4057:9ef4:1e28%6
IPv4-адрес. . . . . : 192.168.1.69
Маска подсети . . . . . : 255.255.255.0
Основной шлюз. . . . . : 192.168.1.1
PS D:\work\emcikunova\vagrant>
PS D:\work\emcikunova\vagrant> ping 192.168.1.1

Обмен пакетами с 192.168.1.1 по с 32 байтами данных:
Ответ от 192.168.1.1: число байт=32 время<1мс TTL=64
Ответ от 192.168.1.1: число байт=32 время=1мс TTL=64
Ответ от 192.168.1.1: число байт=32 время<1мс TTL=64
Ответ от 192.168.1.1: число байт=32 время<1мс TTL=64

Статистика Ping для 192.168.1.1:
    Пакетов: отправлено = 4, получено = 4, потеряно = 0
    (0% потерь)
    Приблизительное время приема-передачи в мс:
        Минимальное = 0мсек, Максимальное = 1 мсек, Среднее = 0 мсек
PS D:\work\emcikunova\vagrant> |
```

Рисунок 1: Просмотр IP-адреса и шлюза по умолчанию

2.2 Определяю MAC-адреса устройств

arp or icmp					
No.	Time	Source	Destination	Protocol	Length Info
26	7.891221	ASUSTekCOMPU_78:b6:f2	Intel_6f:7b:ec	ARP	42 Who has 192.168.1.69? Tell 192.168.1.1
27	7.891250	Intel_6f:7b:ec	ASUSTekCOMPU_78:b6:f2	ARP	42 192.168.1.69 is at f8:fe:5e:6f:7b:ec
102	14.451241	192.168.1.69	192.168.1.1	ICMP	74 Echo (ping) request id=0x0001, seq=33/8448, ttl=128 (reply in 103)
103	14.451904	192.168.1.1	192.168.1.69	ICMP	74 Echo (ping) request id=0x0001, seq=33/8448, ttl=64 (request in 102)
106	15.455314	192.168.1.69	192.168.1.1	ICMP	74 Echo (ping) request id=0x0001, seq=34/8704, ttl=128 (reply in 107)
107	15.456221	192.168.1.1	192.168.1.69	ICMP	74 Echo (ping) reply id=0x0001, seq=34/8704, ttl=64 (request in 106)

> Frame 102: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface v	0000	c8 7f 54 78 b6 f2 f8 fe	5e 6f 7b ec 08 00 45 00	..Tx....^o{...E-
> Ethernet II, Src: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec), Dst: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2)	0010	00 3c f8 90 00 00 00 01	00 00 c0 a8 01 45 c0 a8	<.....E-
Destination: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2)	0020	01 01 08 00 4d 3a 00 01	00 21 61 62 63 64 65 66	...N...!abcdef
.....0..... = L0 bit: Globally unique address (factory default)	0030	67 68 69 6a 6b 6c 6d 6e	6f 70 71 72 73 74 75 76	ghijklm opqrstuv
.....0..... = IG bit: Individual address (unicast)	0040	77 61 62 63 64 65 66 67	68 69	wabcedfg hi
Source: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)				
.....0..... = LG bit: Globally unique address (factory default)				
.....0..... = IG bit: Individual address (unicast)				
Type: IPv4 (0x0800)				
[Stream index: 3]				
> Internet Protocol Version 4, Src: 192.168.1.69, Dst: 192.168.1.1				
0100 = Version: 4				
....0101 = Header Length: 20 bytes (5)				
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)				
Total Length: 60				
Identification: 0xf890 (63632)				
0000 = Flags: 0x0				
...0 0000 0000 0000 = Fragment Offset: 0				
Time to Live: 128				
Protocol: ICMP (1)				
Header Checksum: 0x0000 [validation disabled]				
[Header checksum status: Unverified]				
Source Address: 192.168.1.69				
Destination Address: 192.168.1.1				
[Stream index: 3]				
> Internet Control Message Protocol				
Type: Echo (ping) request (8)				
Code: 0				
Checksum: 0x4d3a [correct]				
[Checksum Status: Good]				
Identifier (BE): 1 (0x0001)				
Identifier (LE): 256 (0x0100)				
Sequence Number (BE): 33 (0x0021)				
Sequence Number (LE): 8448 (0x2100)				
[Response frame: 103]				
> Data (32 bytes)				

Рисунок 2: MAC-адрес компьютера и маршрутизатора

2.3 Анализирую ICMP Echo Reply

arp or icmp						
No.	Time	Source	Destination	Protocol	Length	Info
26	7.891221	ASUSTekCOMPU_78:b6:f2	Intel_6f:7b:ec	ARP	42	Who has 192.168.1.69? Tell 192.168.1.1
27	7.891250	Intel_6f:7b:ec	ASUSTekCOMPU_78:b6:f2	ARP	42	192.168.1.69 is at f8:fe:5e:0f:7b:ec
102	14.451241	192.168.1.69	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=33/8448, ttl=128 (reply in 103)
103	14.451984	192.168.1.1	192.168.1.69	ICMP	74	Echo (ping) reply id=0x0001, seq=33/8448, ttl=64 (request in 102)
106	15.455314	192.168.1.69	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=34/8704, ttl=128 (reply in 107)
107	15.456221	192.168.1.1	192.168.1.69	ICMP	74	Echo (ping) reply id=0x0001, seq=34/8704, ttl=64 (request in 106)
Frame 103: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface V						
Ethernet II, Src: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2), Dst: Intel_6f:7b:ec (f8:fe:5e:0f:7b:ec)						
Destination: Intel_6f:7b:ec (f8:fe:5e:0f:7b:ec)						
.....0..... = LG bit: Globally unique address (factory default)						
.....0..... = IG bit: Individual address (unicast)						
Source: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2)						
.....0..... = LG bit: Globally unique address (factory default)						
.....0..... = IG bit: Individual address (unicast)						
Type: IPv4 (0x0800)						
[Stream index: 3]						
Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.69						
0100 = Version: 4						
....0101 = Header Length: 20 bytes (5)						
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)						
Total Length: 60						
Identification: 0x24e3 (9443)						
0000 = Flags: 0x0						
...0 0000 0000 0000 = Fragment Offset: 0						
Time to Live: 64						
Protocol: ICMP (1)						
Header Checksum: 0xd247 [validation disabled]						
[Header checksum status: Unverified]						
Source Address: 192.168.1.1						
Destination Address: 192.168.1.69						
[Stream index: 3]						
Internet Control Message Protocol						
Type: Echo (ping) reply (0)						
Code: 0						
Checksum: 0x553a [correct]						
[Checksum Status: Good]						
Identifier (BE): 1 (0x0001)						
Identifier (LE): 256 (0x0100)						
Sequence Number (BE): 33 (0x0021)						
Sequence Number (LE): 8448 (0x2100)						
[Request frame: 102]						
[Response time: 0,743 ms]						
Data (32 bytes)						

Рисунок 3: Ответ ICMP от шлюза

2.4 Анализирую ARP-запрос

arp or icmp							
No.	Time	Source	Destination	Protocol	Length	Info	
26	7.891221	ASUSTekCOMPU_78:b6:f2	Intel_6f:7b:ec	ARP	42	Who has 192.168.1.69? Tell 192.168.1.1	
27	7.891250	Intel_6f:7b:ec	ASUSTekCOMPU_78:b6:f2	ARP	42	192.168.1.69 is at f8:fe:5e:6f:7b:ec	
102	14.451241	192.168.1.69	192.168.1.1	ICMP	74	Echo (ping) request id=8x0001, seq=33/8448, ttl=128 (reply in 103)	
103	14.451984	192.168.1.1	192.168.1.69	ICMP	74	Echo (ping) reply id=8x0001, seq=33/8448, ttl=64 (request in 102)	
106	15.455314	192.168.1.69	192.168.1.1	ICMP	74	Echo (ping) request id=8x0001, seq=34/8704, ttl=128 (reply in 107)	
107	15.456221	192.168.1.1	192.168.1.69	ICMP	74	Echo (ping) reply id=8x0001, seq=34/8704, ttl=64 (request in 106)	

> Frame 26: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface D	0000	f8 fe 5e 6f 7b ec c8 7f 54 78 b6 f2 00 06 00 01	...o{... Tx.....
↳ Ethernet II, Src: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2), Dst: Intel_6f:7b:ec (f8:fe:5e:	0010	00 00 06 04 00 01 c8 7f 54 78 b6 f2 c0 a8 01 01	 Tx.....
↳ Destination: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)	0020	00 00 00 00 00 c0 a8 01 45	 E
↳ ..0. = LG bit: Globally unique address (factory default)			
↳ ..0. = IG bit: Individual address (unicast)			
↳ Source: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2)			
↳ ..0. = LG bit: Globally unique address (factory default)			
↳ ..0. = IG bit: Individual address (unicast)			
Type: ARP (0x0806)			
[Stream index: 3]			
↳ Address Resolution Protocol (request)			
Hardware type: Ethernet (1)			
Protocol type: IPv4 (0x0800)			
Hardware size: 6			
Protocol size: 4			
Opcode: request (1)			
Sender MAC address: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2)			
Sender IP address: 192.168.1.1			
Target MAC address: 00:00:00:00:00:00 (00:00:00:00:00:00)			
Target IP address: 192.168.1.69			

Рисунок 4: ARP Request от маршрутизатора

2.5 Анализирую ARP-ответ

arp or icmp

No.	Time	Source	Destination	Protocol	Length	Info
26	7.891221	ASUSTekCOMPU_78:b6:f2:c8:7f:54	Intel_6f:7b:ec	ARP	42	Who has 192.168.1.69? Tell 192.168.1.1
27	7.891250	Intel_6f:7b:ec	ASUSTekCOMPU_78:b6:f2:c8:7f:54	ARP	42	192.168.1.69 is at f8:fe:5e:6f:7b:ec
102	14.451241	192.168.1.69	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=33/8448, ttl=128 (reply in 103)
103	14.451984	192.168.1.1	192.168.1.69	ICMP	74	Echo (ping) reply id=0x0001, seq=33/8448, ttl=64 (request in 102)
106	15.455314	192.168.1.69	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=34/8704, ttl=128 (reply in 107)
107	15.456221	192.168.1.1	192.168.1.69	ICMP	74	Echo (ping) reply id=0x0001, seq=34/8704, ttl=64 (request in 106)
> Frame 27: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF{...}						
Ethernet II, Src: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec), Dst: ASUSTekCOMPU_78:b6:f2:c8:7f:54 (c8:7f:54:b6:f2:c8:7f:54)						
Destination: ASUSTekCOMPU_78:b6:f2:c8:7f:54 (c8:7f:54:b6:f2:c8:7f:54)						
.....0..... = LG bit: Globally unique address (factory default)						
.....0..... = IG bit: Individual address (unicast)						
Source: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)						
.....0..... = LG bit: Globally unique address (factory default)						
.....0..... = IG bit: Individual address (unicast)						
Type: ARP (0x0006)						
[Stream index: 3]						
Address Resolution Protocol (reply)						
Hardware type: Ethernet (1)						
Protocol type: IPv4 (0x0000)						
Hardware size: 6						
Protocol size: 4						
Opcode: reply (2)						
Sender MAC address: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)						
Sender IP address: 192.168.1.69						
Target MAC address: ASUSTekCOMPU_78:b6:f2:c8:7f:54 (c8:7f:54:b6:f2:c8:7f:54)						
Target IP address: 192.168.1.1						

Рисунок 5: ARP Reply от компьютера

2.6 Проверяю доступность удалённого узла

```
PS D:\work\emcikunova\vagrant>
PS D:\work\emcikunova\vagrant> ping www.yandex.ru

Обмен пакетами с www.yandex.ru [77.88.44.55] с 32 байтами данных:
Ответ от 77.88.44.55: число байт=32 время=4мс TTL=55
Ответ от 77.88.44.55: число байт=32 время=4мс TTL=55
Ответ от 77.88.44.55: число байт=32 время=4мс TTL=55
Ответ от 77.88.44.55: число байт=32 время=4мс TTL=55

Статистика Ping для 77.88.44.55:
    Пакетов: отправлено = 4, получено = 4, потеряно = 0
    (0% потерь)

Приблизительное время приема-передачи в мс:
    Минимальное = 4мсек, Максимальное = 4 мсек, Среднее = 4 мсек
PS D:\work\emcikunova\vagrant> |
```

Рисунок 6: Проверка доступности www.yandex.ru

2.7 Изучаю ICMP-запрос к удалённому узлу

arp or icmp						
No.	Time	Source	Destination	Protocol	Length	Info
13	8.822211	192.168.1.69	77.88.44.55	ICMP	74	Echo (ping) request id=0x0001, seq=37/9472, ttl=128 (reply in 14)
14	8.826988	77.88.44.55	192.168.1.69	ICMP	74	Echo (ping) reply id=0x0001, seq=37/9472, ttl=55 (request in 13)
15	9.837922	192.168.1.69	77.88.44.55	ICMP	74	Echo (ping) request id=0x0001, seq=38/9728, ttl=128 (reply in 16)
16	9.842589	77.88.44.55	192.168.1.69	ICMP	74	Echo (ping) reply id=0x0001, seq=38/9728, ttl=55 (request in 15)
18	10.857392	192.168.1.69	77.88.44.55	ICMP	74	Echo (ping) request id=0x0001, seq=39/9984, ttl=128 (reply in 19)
19	10.862078	77.88.44.55	192.168.1.69	ICMP	74	Echo (ping) reply id=0x0001, seq=39/9984, ttl=55 (request in 18)

> Frame 13: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface 10
> Ethernet II, Src: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec), Dst: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2)
> Destination: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2)
.....0..... = LG bit: Globally unique address (factory default)
.....0..... = IG bit: Individual address (unicast)
> Source: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)
.....0..... = LG bit: Globally unique address (factory default)
.....0..... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 3]
> Internet Protocol Version 4, Src: 192.168.1.69, Dst: 77.88.44.55
0100..... = Version: 4
....0101 = Header Length: 20 bytes (5)
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 60
Identification: 0xcfc2c (53036)
> 000..... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: ICMP (1)
Header Checksum: 0x0000 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.1.69
Destination Address: 77.88.44.55
[Stream index: 6]
> Internet Control Message Protocol
Type: Echo (ping) request (8)
Code: 0
Checksum: 0x4d36 [correct]
[Checksum Status: Good]
Identifier (BE): 1 (0x0001)
Identifier (LE): 256 (0x0100)
Sequence Number (BE): 37 (0x0025)
Sequence Number (LE): 9472 (0x2500)
[Response Number: 14]
> Data (32 bytes)

Рисунок 7: ICMP Echo Request к удалённому адресу

2.8 Изучаю ICMP-ответ от удалённого узла

arp or icmp									
No.	Time	Source	Destination	Protocol	Length	Info			
→ 13	8.822211	192.168.1.69	77.88.44.55	ICMP	74	Echo (ping) request id=0x0001, seq=37/9472, ttl=128 (reply in 14)			
← 14	8.826988	77.88.44.55	192.168.1.69	ICMP	74	Echo (ping) reply id=0x0001, seq=37/9472, ttl=55 (request in 13)			
→ 15	9.837922	192.168.1.69	77.88.44.55	ICMP	74	Echo (ping) request id=0x0001, seq=38/9728, ttl=128 (reply in 16)			
→ 16	9.842589	77.88.44.55	192.168.1.69	ICMP	74	Echo (ping) reply id=0x0001, seq=38/9728, ttl=55 (request in 15)			
→ 18	10.857392	192.168.1.69	77.88.44.55	ICMP	74	Echo (ping) request id=0x0001, seq=39/9984, ttl=128 (reply in 19)			
→ 19	10.862078	77.88.44.55	192.168.1.69	ICMP	74	Echo (ping) reply id=0x0001, seq=39/9984, ttl=55 (request in 18)			
Frame 14: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF{...}									
Ethernet II, Src: ASUSTekCOMPU_78:b6:f2 (c8:f7:54:78:b6:f2), Dst: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)									
Destination: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)									
.....0..... = LG bit: Globally unique address (factory default)									
.....0..... = IG bit: Individual address (unicast)									
Source: ASUSTekCOMPU_78:b6:f2 (c8:f7:54:78:b6:f2)									
.....0..... = LG bit: Globally unique address (factory default)									
.....0..... = IG bit: Individual address (unicast)									
Type: IPv4 (0x0000)									
[Stream Index: 3]									
Internet Protocol Version 4, Src: 77.88.44.55, Dst: 192.168.1.69									
0100 = Version: 4									
... 0101 = Header Length: 20 bytes (5)									
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)									
Total Length: 60									
Identification: 0xcfc2c (53036)									
0000 = Flags: 0x0									
...0 0000 0000 0000 = Fragment Offset: 0									
Time to Live: 55									
Protocol: ICMP (1)									
Header Checksum: 0x7918 [validation disabled]									
[Header checksum status: Unverified]									
Source Address: 77.88.44.55									
Destination Address: 192.168.1.69									
[Stream index: 6]									
Internet Control Message Protocol									
Type: Echo (ping) reply (0)									
Code: 0									
Checksum: 0x5536 [correct]									
[Checksum Status: Good]									
Identifier (BE): 1 (0x0001)									
Identifier (LE): 256 (0x0100)									
Sequence Number (BE): 37 (0x0025)									
Sequence Number (LE): 9472 (0x2500)									
[Request ID: 13]									
[Response time: 4,777 ms]									
Data (32 bytes)									

Рисунок 8: ICMP Echo Reply от удалённого адреса

2.9 Анализирую HTTP-запрос

No.	Time	Source	Destination	Protocol	Length	Info
335	44.817251	192.168.1.69	188.184.67.127	HTTP	342	GET /apple-touch-icon.png HTTP/1.1
339	44.850014	192.168.1.69	188.184.67.127	HTTP	497	GET / HTTP/1.1
341	44.988498	188.184.67.127	192.168.1.69	HTTP	932	HTTP/1.1 200 OK (text/html)
340	44.988895	192.168.1.69	188.184.67.127	HTTP	438	GET /favicon.ico HTTP/1.1
358	45.838584	188.184.67.127	192.168.1.69	HTTP	248	HTTP/1.1 200 OK (image/vnd.microsoft.icon)
375	49.713200	192.168.1.69	188.184.67.127	HTTP	557	GET /hypertext/WWW/TheProject.html HTTP/1.1
381	49.771161	188.184.67.127	192.168.1.69	HTTP	1044	HTTP/1.1 200 OK (text/html)
411	57.196182	188.184.67.127	192.168.1.69	HTTP	414	[TCP Retransmission] HTTP/1.1 404 Not Found (text/html)

> Internet Protocol Version 4, Src: 192.168.1.69, Dst: 188.184.67.127	0000 c8 7f 54 78 b6 f2 f8 fe 5e 6f 7b ec 08 00 45 00 --Tx
Transmission Control Protocol, Src Port: 52753, Dst Port: 80, Seq: 1, Ack: 1, Len: 503	0010 02 1f be d8 40 00 88 06 00 00 c8 a8 01 45 bc b8 ---@
Source Port: 52753	0020 43 7f ce 11 00 50 2c 84 ed e9 31 45 7f 0c 50 18 C---
Destination Port: 80	0030 02 01 c4 36 00 00 47 45 54 20 2f 68 79 70 65 72 ---6-
[Stream index: 18]	0040 74 65 70 74 2f 57 57 57 2f 54 68 65 50 72 6f 6a text/
[Stream Packet Number: 4]	0050 65 63 74 2e 68 74 6d 6c 20 48 54 54 50 2f 31 2e ect.h
[Conversation completeness: Complete, WITH_DATA (31)]	0060 31 0d 0a 48 6f 73 74 3a 20 69 6e 66 6f 2e 63 65 1--Ho
[TCP Segment Len: 503]	0070 72 6e 2e 63 68 0d 0a 43 6f 6e 6e 65 63 74 69 6f rn.ch
Sequence Number: 1 (relative sequence number)	0080 6e 3a 20 6b 65 65 70 2d 61 6c 69 76 65 6d 0a 55 n: ke
Sequence Number (raw): 746989161	0090 70 67 72 61 64 65 2d 49 6e 73 65 63 75 72 65 2d pgrad
[Next Sequence Number: 504 (relative sequence number)]	00a0 52 65 71 75 65 73 74 73 3a 20 31 0d 0a 55 73 65 Reque
Acknowledgment Number: 1 (relative ack number)	00b0 72 2d 41 67 65 6e 74 3a 20 4d 6f 7a 69 6c 6c 61 r-Age
Acknowledgment number (raw): 826638092	00c0 2f 35 2e 38 20 28 57 69 6e 64 6f 77 73 20 4e 54 /5.8
0101 ... = Header Length: 20 bytes (5)	00d0 20 31 30 2e 30 3b 20 57 69 6e 36 34 3b 20 70 36 18.0
Flags: 0x018 (PSH, ACK)	00e0 34 29 20 41 70 70 6c 65 57 65 62 4b 69 74 2f 35 4) Ap
Window: 513	00f0 33 37 2e 33 36 20 28 4b 48 54 4d 4c 2c 20 6c 69 37.36
[Calculated window size: 131328]	0100 6b 65 20 47 65 63 6b 6f 29 20 43 68 72 6f 6d 65 ke Ge
[Window size scaling factor: 256]	0110 2f 31 35 30 2e 30 2e 30 2e 30 20 59 61 42 72 6f /150.
Checksum: 0xc436 [unverified]	0120 77 73 65 72 2f 32 36 2e 30 2e 30 2e 30 20 53 61 wser/
[Checksum Status: Unverified]	0130 66 61 72 69 2f 35 33 37 2e 33 36 0d 0a 41 63 63 far:/
Urgent Pointer: 0	0140 65 70 74 3a 20 74 65 78 74 2f 68 74 6d 6c 2c 61 ept:
[Timestamps]	0150 70 70 6c 69 63 61 74 69 6f 6e 2f 7b 68 74 6d 6c pplic
[SEQ/ACK analysis]	0160 2b 78 6d 6c 2c 61 70 70 6c 69 63 61 74 69 6f 6e xml,
[Client Contiguous Streams: 1]	0170 2f 78 6d 6c 3b 71 3d 30 2e 39 2c 69 6d 61 67 65 /xml;
[Server Contiguous Streams: 1]	0180 2f 61 76 69 66 2c 69 6d 61 67 65 2f 77 65 62 70 /avif
TCP payload (503 bytes)	0190 2c 69 6d 61 67 65 2f 61 70 6e 67 2c 2a 2f 2a 3b ,img
> Hypertext Transfer Protocol	01a0 71 3d 30 2e 38 2c 61 70 70 6c 69 63 61 74 69 6f q=0.8
> GET /hypertext/WWW/TheProject.html HTTP/1.1\r\n	01b0 6e 2f 73 69 67 6e 65 64 2d 65 78 63 68 61 6e 67 n/sig
Host: info.cern.ch\r\n	01c0 65 3b 76 3d 62 33 3b 71 3d 30 2e 37 0d 0a 52 65 ejvb=
Connection: keep-alive\r\n	01d0 66 72 65 72 3a 20 68 74 7a 70 3a 2f 2f 69 6e ferer
Upgrade-Insecure-Requests: 1\r\n	01e0 66 6f 2e 63 65 72 6e 2c 63 68 2f 0d 0a 41 63 63 fo.ce
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like	01f0 65 70 74 2d 45 6e 63 6f 64 69 6e 67 3a 20 67 7a ept-E
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,in	0200 69 70 2c 20 64 65 66 6c 61 74 65 0d 0a 41 63 63 ip, d
Referer: http://info.cern.ch/r\n	0210 65 70 74 2d 4c 61 6e 67 75 61 67 65 3a 20 72 75 ept-L
Accept-Encoding: gzip, deflate\r\n	0220 2c 65 6e 3b 71 3d 30 2e 39 0d 0a 0a 0a 0a ,enq
Accept-Language: ru,en;q=0.9\r\n	
\r\n	
[Response in frame: 381]	
[Full request URI: http://info.cern.ch/hypertext/WWW/TheProject.html]	

Рисунок 9: HTTP GET-запрос и TCP-сегмент

2.10 Анализирую HTTP-ответ

No.	Time	Source	Destination	Protocol	Length	Info
333	44.017251	192.168.1.69	188.184.67.127	HTTP	342	GET /apple-touch-icon.png HTTP/1.1
339	44.850014	192.168.1.69	188.184.67.127	HTTP	497	GET / HTTP/1.1
341	44.980498	188.184.67.127	192.168.1.69	HTTP	932	HTTP/1.1 200 OK (text/html)
349	44.980885	192.168.1.69	188.184.67.127	HTTP	438	GET /favicon.ico HTTP/1.1
358	45.038584	188.184.67.127	192.168.1.69	HTTP	248	HTTP/1.1 200 OK (image/vnd.microsoft.icon)
375	49.713200	192.168.1.69	188.184.67.127	HTTP	957	GET /hypertext/Adm/TT/Tree/sect.html HTTP/1.1
381	49.771161	188.184.67.127	192.168.1.69	HTTP	1044	HTTP/1.1 200 OK (text/html)
411	57.196182	188.184.67.127	192.168.1.69	HTTP	414	[TCP Retransmission] HTTP/1.1 404 Not Found (text/html)

Internet Protocol Version 4, Src: 188.184.67.127, Dst: 192.168.1.69	
Source Port: 80	
Destination Port: 52753	
[Stream Index: 18]	
[Stream Packet Number: 7]	
[Conversation completeness: Complete, WITH_DATA (31)]	
[TCP Segment Len: 990]	
Sequence Number: 1461 (relative sequence number)	
Sequence Number (raw): 826639552	
[Next Sequence Number: 2451 (relative sequence number)]	
Acknowledgment Number: 504 (relative ack number)	
Acknowledgment number (raw): 746989664	
0101 = Header Length: 20 bytes (5)	
[Flags: 0x018 (PSH, ACK)]	
Window: 501	
[Calculated window size: 64128]	
[Window size scaling factor: 128]	
Checksum: 0xc3e6 [unverified]	
[Checksum Status: Unverified]	
Urgent Pointer: 0	
[Timestamps]	
[SND/ACK analysis]	
[Client Contiguous Streams: 1]	
[Server Contiguous Streams: 1]	
TCP payload (990 bytes)	
TCP segment data (990 bytes)	
[2 Reassembled TCP Segments (2450 bytes): #380(1460), #381(990)]	
Hypertext Transfer Protocol	
Date: Wed, 23 Sep 2020 10:13:36 GMT\r\n	
Server: Apache\r\n	
Last-Modified: Thu, 03 Dec 1992 08:37:20 GMT\r\n	
ETag: \"8a9-291e721905000\"\r\n	
Accept-Ranges: bytes\r\n	
Content-Length: 2217\r\n	
Connection: close\r\n	
Content-Type: text/html\r\n	
\r\n	

Рисунок 10: HTTP-ответ сервера и TCP-сегменты

2.11 Анализирую DNS-запрос

The screenshot displays a Wireshark capture of a DNS transaction. The packet list at the top shows several DNS packets. Packet 368 is selected, showing a standard query response from 192.168.1.69 to 192.168.1.1. The packet details pane shows the User Datagram Protocol (UDP) and Domain Name System (DNS) sections. The DNS section shows a standard query response with transaction ID 0x9490. The packet bytes pane shows the raw data of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
310	43.948547	192.168.1.1	192.168.1.69	DNS	130	Standard query response 0xa03f HTTPS api.browser.yandex.ru SOA ns3.yan
367	46.797897	192.168.1.69	192.168.1.1	DNS	77	Standard query 0x7d3a HTTPS line-mode.cern.ch
368	46.798092	192.168.1.69	192.168.1.1	DNS	77	Standard query 0x9490 A line-mode.cern.ch
369	46.924666	192.168.1.1	192.168.1.69	DNS	174	Standard query response 0x7d3a HTTPS line-mode.cern.ch CNAME lb-paas-1
370	46.931182	192.168.1.1	192.168.1.69	DNS	163	Standard query response 0x9490 A line-mode.cern.ch CNAME lb-paas-line-i
414	57.198265	192.168.1.69	192.168.1.1	DNS	78	Standard query 0xd584 HTTPS favicon.yandex.net
415	57.198450	192.168.1.69	192.168.1.1	DNS	78	Standard query 0xab3d A favicon.yandex.net
416	57.202343	192.168.1.1	192.168.1.69	DNS	142	Standard query response 0xab3d A favicon.yandex.net A 87.250.250.36 A !

> Frame 368: Packet, 77 bytes on wire (616 bits), 77 bytes captured (616 bits) on interface \

> Ethernet II, Src: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec), Dst: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:

> Internet Protocol Version 4, Src: 192.168.1.69, Dst: 192.168.1.1

▼ User Datagram Protocol, Src Port: 64307, Dst Port: 53

- Source Port: 64307
- Destination Port: 53
- Length: 43
- Checksum: 0x83d3 [unverified]
- [Checksum Status: Unverified]
- [Stream index: 28]
- [Stream Packet Number: 1]
- > [Timestamps]
- UDP payload (35 bytes)

▼ Domain Name System (query)

- Transaction ID: 0x9490
- > Flags: 0x0100 Standard query
- Questions: 1
- Answer RRs: 0
- Authority RRs: 0
- Additional RRs: 0
- > Queries
- [Response In: 370]

0000 c8 7f 54 78 b6 f2 f8 fe 5e 6f 7b ec 08 00 45 00

0010 00 3f f8 aa 00 00 00 11 00 00 c0 a8 01 45 c0 a8

0020 01 01 fb 33 00 35 00 2b 83 d3 94 90 01 00 00 01

0030 00 00 00 00 00 00 09 6c 69 6e 65 2d 6d 6f 64 65

0040 04 63 65 72 6e 02 63 68 00 00 01 00 01

Рисунок 11: DNS-запрос по UDP

2.12 Анализирую DNS-ответ

dns						
No.	Time	Source	Destination	Protocol	Length	Info
310	43.948547	192.168.1.1	192.168.1.69	DNS	130	Standard query response 0xa03f HTTPS api.browser.yandex.ru SOA ns3.yandex.ru
367	46.797897	192.168.1.69	192.168.1.1	DNS	77	Standard query 0x7d3a HTTPS line-mode.cern.ch
368	46.798092	192.168.1.69	192.168.1.1	DNS	77	Standard query 0x9490 A line-mode.cern.ch
369	46.924666	192.168.1.1	192.168.1.69	DNS	174	Standard query response 0x7d3a HTTPS line-mode.cern.ch CNAME lb-paas-line-mo
370	46.931182	192.168.1.1	192.168.1.69	DNS	163	Standard query response 0x9490 A line-mode.cern.ch CNAME lb-paas-line-mode-4
414	57.198265	192.168.1.69	192.168.1.1	DNS	78	Standard query 0xd584 HTTPS favicon.yandex.net
415	57.198450	192.168.1.69	192.168.1.1	DNS	78	Standard query 0xab3d A favicon.yandex.net
416	57.202343	192.168.1.1	192.168.1.69	DNS	142	Standard query response 0xab3d A favicon.yandex.net A 87.250.250.36 A 93.158

> Frame 370: Packet, 163 bytes on wire (1304 bits), 163 bytes captured (1304 bits) on interfa	0000	f8 fe 5e 6f 7b ec c8 7f 54 78 b6 f2 08 00 45 00	...
> Ethernet II, Src: ASUSTekCOMPU 78:b6:f2 (c8:7f:54:78:b6:f2), Dst: Intel_6f:7b:ec (f8:fe:5e:	0010	00 95 71 1c 40 00 40 11 45 a5 c0 a8 01 01 c0 a8	...
> Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.69	0020	01 45 00 35 fb 33 00 81 39 85 94 90 81 80 00 01	...
> User Datagram Protocol, Src Port: 53, Dst Port: 64307	0030	00 04 00 00 00 00 09 6c 69 6e 65 2d 6d 6f 64 65	...
Source Port: 53	0040	04 63 65 72 6e 02 63 68 00 00 01 00 01 c0 0c 00	...
Destination Port: 64307	0050	05 00 01 00 00 03 84 00 1a 17 6c 62 2d 70 61 61	...
Length: 129	0060	73 2d 6c 69 6e 65 2d 6d 6f 64 65 2d 34 35 30 34	...
Checksum: 0x3985 [unverified]	0070	37 c0 16 c0 2f 00 01 00 01 00 00 00 3c 00 04 bc	...
[Checksum Status: Unverified]	0080	b9 2a 77 c0 2f 00 01 00 01 00 00 00 3c 00 04 bc	...
[Stream index: 28]	0090	b9 30 28 c0 2f 00 01 00 01 00 00 00 3c 00 04 bc	...
[Stream Packet Number: 2]	00a0	b9 2e 79	...
[Timestamps]			
UDP payload (121 bytes)			
> Domain Name System (response)			
Transaction ID: 0x9490			
> Flags: 0xB180 Standard query response, No error			
Questions: 1			
Answer RRs: 4			
Authority RRs: 0			
Additional RRs: 0			
> Queries			
> Answers			
[Request In: 368]			
[Time: 133.090000 milliseconds]			

Рисунок 12: DNS-ответ по UDP

2.13 Проверяю наличие QUIC-трафика

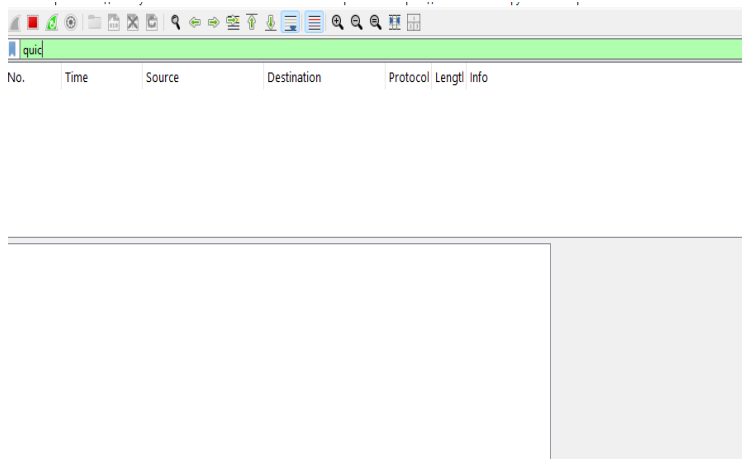


Рисунок 13: Отсутствие QUIC-пакетов в захвате

2.14 Анализирую TLS поверх TCP

Примените фильтр отображения ... «Ctrl-/»						
No.	Time	Source	Destination	Protocol	Length	Info
968	313.776625	87.250.251.183	192.168.1.69	TCP	54	443 → 58526 [ACK] Seq=1 Ack=1411 Win=41728 Len=0
969	313.776625	87.250.251.183	192.168.1.69	TCP	54	443 → 58526 [ACK] Seq=1 Ack=2060 Win=41728 Len=0
970	313.776952	87.250.251.183	192.168.1.69	TLSv1.2	210	Server Hello, Change Cipher Spec, Encrypted Handshake Message
971	313.777221	192.168.1.69	87.250.251.183	TLSv1.2	105	Change Cipher Spec, Encrypted Handshake Message
972	313.777421	192.168.1.69	87.250.251.183	TLSv1.2	1167	Application Data
973	313.777452	192.168.1.69	87.250.251.183	TLSv1.2	952	Application Data
974	313.782996	87.250.251.183	192.168.1.69	TCP	54	443 → 58526 [ACK] Seq=157 Ack=4122 Win=42496 Len=0
975	313.895581	87.250.251.183	192.168.1.69	TCP	1274	443 → 58526 [ACK] Seq=157 Ack=4122 Win=42496 Len=1220 [TCP PDU reassembled]
Sequence Number: 1 (relative sequence number)						
Sequence Number (raw): 92288597						
[Next Sequence Number: 157 (relative sequence number)]						
Acknowledgment Number: 2060 (relative ack number)						
Acknowledgment number (raw): 1471728672						
0101 ... = Header Length: 20 bytes (5)						
Flags: 0x018 (PSH, ACK)						
Window: 166						
[Calculated window size: 42496]						
[Window size scaling factor: 256]						
Checksum: 0x57c5 [unverified]						
[Checksum Status: Unverified]						
Urgent Pointer: 0						
[Timestamps]						
[SEQ/ACK analysis]						
[Client Contiguous Streams: 1]						
[Server Contiguous Streams: 1]						
TCP payload (156 bytes)						
Transport Layer Security						
[Stream index: 18]						
TLSv1.2 Record Layer: Handshake Protocol: Server Hello						
Content Type: Handshake (22)						
Version: TLS 1.2 (0x0303)						
Length: 100						
Handshake Protocol: Server Hello						
TLSv1.2 Record Layer: Change Cipher Spec Protocol: Change Cipher Spec						
Content Type: Change Cipher Spec (20)						
Version: TLS 1.2 (0x0303)						
Length: 1						
Change Cipher Spec Message						
[Expert Info (Note/Sequence): This session reuses previously negotiated keys (Se						
[This session reuses previously negotiated keys (Session resumption)]						
[Severity level: Note]						
[Group: Sequence]						
TLSv1.2 Record Layer: Handshake Protocol: Encrypted Handshake Message						
Content Type: Handshake (22)						
Version: TLS 1.2 (0x0303)						
Length: 40						
Handshake Protocol: Encrypted Handshake Message						

Рисунок 14: Передача TLS 1.2 поверх TCP

2.15 Анализирую TCP handshake

No.	Time	Source	Destination	Protocol	Length	Info
14	2.395839	192.168.1.69	188.184.67.127	HTTP	669	GET /hypertext/WWW/TheProject.html HTTP/1.1
15	2.399069	188.184.67.127	192.168.1.69	TCP	66	80 → 54245 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
16	2.399130	192.168.1.69	188.184.67.127	TCP	54	54245 → 80 [ACK] Seq=1 Ack=1 Win=131328 Len=0
17	2.451009	188.184.67.127	192.168.1.69	TCP	54	80 → 50790 [ACK] Seq=1 Ack=616 Win=63744 Len=0
18	2.451054	188.184.67.127	192.168.1.69	HTTP	250	HTTP/1.1 304 Not Modified
19	2.452333	192.168.1.69	188.184.67.127	TCP	54	50790 → 80 [FIN, ACK] Seq=616 Ack=197 Win=131072 Len=0
20	2.453070	188.184.67.127	192.168.1.69	TCP	54	80 → 50790 [FIN, ACK] Seq=197 Ack=616 Win=63744 Len=0
21	2.453130	192.168.1.69	188.184.67.127	TCP	54	50790 → 80 [ACK] Seq=617 Ack=198 Win=131072 Len=0
24	2.460919	192.168.1.69	87.250.251.20	TLSv1.2	169	Application Data
25	2.460940	192.168.1.69	87.250.251.20	TLSv1.2	100	Application Data
26	2.460955	192.168.1.69	87.250.251.20	TLSv1.2	1116	Application Data

Destination Port: 54245
[Stream index: 2]
[Stream Packet Number: 2]
> [Conversation completeness: Incomplete, ESTABLISHED (7)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 3497790773
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 210020935
1000 = Header Length: 32 bytes (8)
▼ **Flags: 0x012 (SYN, ACK)**
000. = Reserved: Not set
...0 = Accurate ECN: Not set
....0 = Congestion Window Reduced: Not set
....0... = ECH-Echo: Not set
....0. = Urgent: Not set
....1 = Acknowledgment: Set
....0... = Push: Not set
....0... = Reset: Not set
▼ **xxxx...x...1 = Syn: set**
▼ [Expert Info (Chat/Sequence): Connection establish acknowledge (SYN+ACK): server
[Connection establish acknowledge (SYN+ACK): server port 80]
[Severity Level: Chat]
[Group: Sequence]
....0... = Fin: Not set
[TCP Flags:A..S.]
Window: 64240
[Calculated window size: 64240]
Checksum: 0x0C39 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
> Options: (12 bytes), Maximum segment size, No-Operation (NOP), No-Operation (NOP), SACK
> [Timestamps]
> [SEQ/ACK analysis]
[Client Contiguous Streams: 1]

0000 f8 fe 5e 6f 7b ec c8 7f 54 78 b6 f2 00 00 45 00 --o
0010 00 34 00 00 40 00 30 06 85 9f bc b8 43 7f c8 a0 -4-
0020 01 45 00 50 d3 e5 d0 7c 0d 35 7d 87 76 43 80 12 -P
0030 fa f0 0c 39 00 00 02 04 05 b4 01 01 04 02 01 03 --9
0040 03 07 ..

Рисунок 15: Трёхэтапное установление TCP-соединения

2.16 Строю график потока TCP

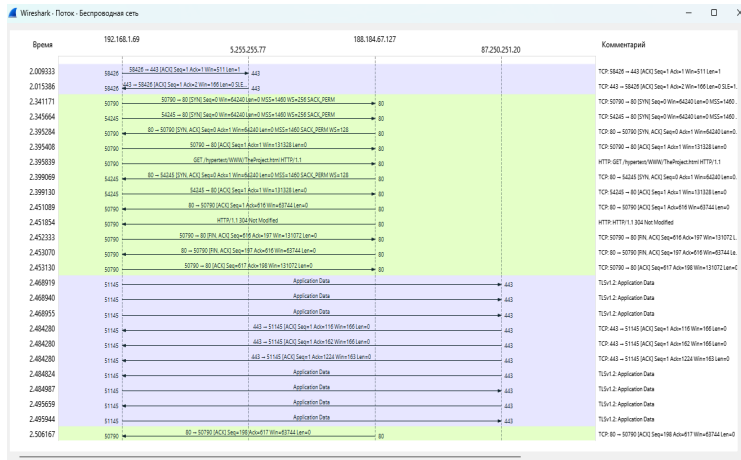


Рисунок 16: График потока TCP-соединения

В ходе лабораторной работы были изучены MAC-адреса сетевых интерфейсов и их структура.

С помощью Wireshark были проанализированы кадры ARP и ICMP. Было установлено, что при обращении к удалённому узлу кадры Ethernet передаются на MAC-адрес шлюза по умолчанию.

Также были рассмотрены протоколы HTTP, DNS и TLS. HTTP использовал TCP, DNS-запросы передавались по UDP, а QUIC-трафик в захвате отсутствовал.

Отдельно был изучен процесс установки TCP-соединения. Handshake выполнялся по схеме **SYN → SYN, ACK → ACK**, после чего начиналась передача прикладных данных.